

Doohickey blueprint

Sirusdog

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1 First run & setup

1.1 Initial setup

1. Flash an SD card with Rasbian Lite using the raspberry pi imager. Keep in mind that this code should work on other hardware/os versions, but setup will likely differ.
2. Plug in the RPI, connect it to a display via HDMI and give it power via the USB-C port.
3. Connect the RPI to the internet, either via it's ethernet port or WIFI.

1.2 Software installs and interface setup

1. Run `sudo apt update` and `sudo apt full-upgrade`
2. Run `sudo apt install git` (i forgot if git is installed by default)
3. Run `git clone https://github.com/Sirusdog/Thermal-camera-project.git`
4. Run `cd Thermal-Camera-Project`
5. Run `pip install -r requirements.txt --break-system-packages`
6. Run `sudo raspi-config`
 - (a) Select *3 Interface Options*
 - (b) Enable *I4 SPI*
 - (c) Enable *I5 I2C*
 - (d) Select *I6 Serial port*
 - i. Select *No* for "Would you like a login shell to be accessible over serial?"
 - ii. Select *Yes* for "Would you like the serial port hardware to be enabled?"
 - (e) Optionally you can disable the sudo password and enable auto-login to root. I'd also recommend enabling SSH.
 - (f) Apply the settings.
7. Run `sudo reboot`

1.3 Cythonizing main and setting up auto-run

1. Run `cd Thermal-Camera-Project`
2. Run `python3 initialiser.py`
3. Run `python setup.py build_ext --inplace`
4. Run `sudo nano thermalCamBoot.service`
 - (a) Change the line `ExecStart=/user.bin/python3 /home/sirus/Thermal-camera-project/run.py` so that the path matches the username of your raspi, i.e if you set the pi's username to pi it'll be `ExecStart=/user.bin/python3 /home/pi/Thermal-camera-project/run.py`
 - (b) Press `Ctrl + X` and then `Y` to save the file.
5. Run `sudo cp thermalCamBoot.service /etc/systemd/system/`
6. Run `sudo systemctl daemon-reload`
7. Run `sudo systemctl enable thermalCamBoot` TODO: Check if this starts the program.

Past this point I'd recommend assembling the (using a normal screen and psu) to test if everything software-wise is working.

1.4 Setting up composite display

1. Run `sudo raspi-config`
2. Select *2 Display Options*
3. Select *D4 Composite*
4. Select *Yes* for "Would you like composite video output to be enabled?"
5. Press *Ok, Finish*
6. Run `sudo reboot`
7. Connect the display to the composite output.
8. Verify that it's working before moving on. At this stage if there's no display output you'll need to SSH into the raspi to figure out what's happening.

2 Parts

Optical elements

Item ID	Description	Designations
LB1757	F=30 mm, 25.4 mm biconvex lens	f_1
ME1-G01	25.4 mm first surface mirror	m
EBS1	25.4 mm 50:50 Beamsplitter	b
PL25mm	25mm Plossl eyepiece	f_2
LB—	F=20 mm, 12.4 mm biconvex lens	f_3

GX16 Connector Pinout

Pin #	Name	RPI	NVG
1	5v	GPIO 5V pins	Thermal camera 5V in, display 5V in
2	GND 1	GPIO GND pin	Thermal camera GND, rotary encoder GND
3	3.3v	GPIO 3.3V pins	Rotary encoder VCC
4	Encoder CLK	GPIO 23	Rotary encoder CLK pin
5	Encoder DT	GPIO 24	Rotary encoder DT pin
6	Encoder SW	GPIO 25	Rotary encoder SW pin
7	Cam CVBS	CVBS-USB board CVBS input	Camera CVBS output
8	Cam RX	UART0 TX	Cam RX pin
9	Cam TX	UART0 RX	Cam TX pin
10	Screen CVBS	RPI composite out	Screen composite in
11	GND 2	GPIO GND pin	Screen GND, brightness controls
12	NC	NC	NC